



Iligan Medical Center College
San Miguel Village, Pala-o, Iligan City
College of Computer Studies



Activity #2 - IAS102
1st Semester, SY 2026 - 2027

Locks, Levels, and Looking In

Bell-LaPadula isn't just theory, it's a set of rules that real operating systems either enforce or quietly fail to enforce, depending on how they're configured. In this activity, you will build a small system with three levels of secrecy inside your own Kali Linux VM, then test it yourself to see whether Linux actually follows those rules on its own, or whether that protection only exists when someone sets it up correctly.

Before You Start

If this is your first time using Kali Linux, this guide explains every click and every command, not just the commands themselves. Nothing here will damage your laptop or your VM. Everything happens inside your Kali virtual machine, which is a separate, contained computer running inside your real one. If something goes wrong, the worst case is restarting the VM.

Go slowly. Type commands exactly as written, including spaces, dashes, and slashes. Linux is case-sensitive and picky about exact spelling. If a command doesn't work, re-check your spelling before asking for help.

Key Terms to Understand

Term	What It Means
VM (Virtual Machine)	<i>A full computer that runs inside your real computer, like a window into another machine. Your Kali VM is separate from your actual laptop.</i>
Terminal	<i>A window where you type text commands instead of clicking icons. This is the main way you'll interact with Kali.</i>
sudo	<i>Short for "super user do." It temporarily gives you admin-level power to run a command. Kali will usually ask for your password when you use it.</i>
User	<i>An account on the system, like a login. In Part 1, you'll create three separate user accounts to represent three levels of secrecy.</i>
Permission	<i>A rule that decides who is allowed to read or write (edit) a file. This is what Bell-LaPadula is really about.</i>



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IP Address	<i>A number that identifies a device on a network, like a home address, but for computers.</i>
Port	<i>A numbered "door" on a computer that a specific service uses to communicate. Port 22, for example, is commonly used for remote login (SSH).</i>
Nmap	<i>A tool that checks which ports are open on a computer. It's the first step in figuring out what's running on a system and what might be vulnerable.</i>

Using Kali Linux for the First Time

Before jumping into the activity, take a few minutes to just look around. Knowing where things are will save you time and confusion later.

1. Starting the VM

- *Open VirtualBox on your laptop, then click on your Kali VM in the left-hand list, then click the green "Start" arrow at the top.*
- *Wait for Kali to finish booting. This can take a minute or two, don't click anything while it loads.*
- *At the login screen, type kali as the username, press Enter, then type kali as the password, press Enter again.*

2. Getting to Know the Desktop

- *Once you're logged in, you'll see the Kali desktop. It looks different from Windows, but the same basic ideas apply. Just take note of this:*
 - *Terminal icon: looks like a small dark rectangle with a >_ symbol inside it. This is the single most important icon for this entire course.*

3. Opening and Using the Terminal

- *Click the black terminal icon in the taskbar to open a terminal window.*
 - *A black window with white or green text should now be open. This is where you'll type every command in this activity, followed by pressing Enter to run it.*

NOTE: If a command asks for a password

When you type a command that starts with sudo, Kali will ask for your password. Type your password and press Enter. Nothing will appear on screen while you type the password, not even dots. That's normal, just type it and press Enter.



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4. Shutting Down Properly

- When you're done for the day, don't just close the VirtualBox window. In the Kali desktop, use the power/shutdown option in the top menu to shut Kali down properly, then close VirtualBox. This avoids corrupting your VM's saved state.

Before You Start the Activity

Go through this checklist before you begin Part 1. This takes less than five minutes and prevents most of the problems that eat up lab time.

Pre-Activity Checklist

- My Kali VM is installed and boots successfully.
- I can see and open the terminal.
- VirtualBox Guest Additions is installed
 - (My screen resizes properly, and I can copy-paste between my laptop and the VM).
- I have taken a snapshot of my VM in its current, working state
 - (VirtualBox menu: Machine > Take Snapshot). This lets me roll back instantly if something breaks.

Why take a snapshot?

- A snapshot saves your VM's exact current state. If anything in this activity goes wrong, like a mistyped command that breaks something, you can restore the snapshot and start over in seconds instead of reinstalling Kali from scratch.



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Part 1: Simulating Bell-LaPadula with Linux Permissions

In this part, you'll create three pretend "employees" at three different secrecy levels, then test whether Kali actually enforces Bell-LaPadula's rules on its own, or whether that only happens if someone sets it up correctly.

Step 1: Create three user accounts.

These three accounts represent Public, Confidential, and Secret levels of secrecy, the same three layers from your Ana example in lecture.

```
sudo useradd -m ana_public
sudo useradd -m ana_confidential
sudo useradd -m ana_secret
```

What you should see: Nothing prints on the screen if it worked. In Linux, no news is good news, if there's no error message, the command succeeded.

- **If you need to start over**

- If you make a mistake and want to redo Part 1 from scratch, re-running useradd on a name that already exists will fail with "user already exists."
- To remove a user and start clean, run `sudo userdel -r ana_public` (repeat for the other two names as needed), then start again from Step 1.

Step 2: Create three folders and assign each to its corresponding user.

This step creates the actual "rooms" (folders) for each secrecy level and ensures that only the user at that level can normally access their own room.

```
sudo mkdir -p /levels/public /levels/confidential /levels/secret
sudo chown ana_public:ana_public /levels/public
sudo chown ana_confidential:ana_confidential /levels/confidential
sudo chown ana_secret:ana_secret /levels/secret
sudo chmod 700 /levels/public /levels/confidential /levels/secret
```

What do chown and chmod mean?

- `chown` changes who owns a folder ("change owner"). `chmod` changes the permission rules on it ("change mode"). The number 700 means: the owner can read/write/open it, and absolutely no one else can, not even other regular users.

What you should see: Again, no output means success. If you see "Permission denied," you likely forgot `sudo` at the start of the line.



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Let's actually see the permission rule we just set, not just take it on faith.

```
ls -l /levels
```

What you should see: A line for each folder, starting with something like drwx----- followed by the owner's name. The drwx----- part is the 700 setting from before, but written out: the owner (rwx) can read, write, and open it, and the two blank groups after it mean no one else can.

This is Bell-LaPadula's rule, made visible.

Step 3: Put a sample file inside each folder.

```
echo "Public info" | sudo tee /levels/public/file.txt
echo "Confidential info" | sudo tee /levels/confidential/file.txt
echo "Secret info" | sudo tee /levels/secret/file.txt
```

What you should see: Each line should print the text back to you, for example Public info, confirming the file was created and written.

Step 4: Switch into the Confidential user's account.

This puts you "in Ana's shoes" for the rest of Part 1. She has Confidential clearance, the middle layer.

```
sudo su - ana_confidential
```

What does su mean?

- su means "switch user." After running this, your terminal is now acting as ana_confidential instead of your normal account until you type exit.
- **Why sudo su and not just su?**
 - su alone asks for the password of the account you're switching into, but ana_confidential doesn't have a password set, so plain su will fail with "Authentication failure." Adding sudo in front means the switch is approved using your own admin rights instead, which is why sudo has to come first.



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Let's confirm the switch actually worked before moving on.

```
whoami
```

- **What you should see:** It should print ana_confidential, not your own username. If it still shows your normal account, the su command didn't work. Scroll up and check for an error message.

Step 5: Try reading all three files.

```
cat /levels/public/file.txt  
cat /levels/confidential/file.txt  
cat /levels/secret/file.txt
```

- cat means "show me what's inside this file." Watch closely: one of these three should fail with "Permission denied." Which one, and does that match what Bell-LaPadula predicts?

Step 6: Try writing to all three files.

```
echo "leak" >> /levels/public/file.txt  
echo "note" >> /levels/confidential/file.txt  
echo "note" >> /levels/secret/file.txt
```

- The >> symbol means "add this text to the end of the file." Again, watch for which one(s) fail.

Go to your Answer Sheet now and fill in the Part 1 Results table before continuing to Step 7.

Step 7: Break the lab on purpose.

Now, let's see what happens if someone configures the system carelessly. First, leave Ana's account and go back to being yourself:

```
exit
```

Now loosen the Public folder's permissions so anyone can write to it:

```
sudo chmod 777 /levels/public
```

What does 777 mean?



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- 777 means everyone, the owner, the group, and all other users, can read, write, and open this folder. It's the opposite of the strict 700 setting from Step 2.

Now switch back into Ana's account and try the write-down action again:

```
sudo su - ana_confidential
echo "leak" >> /levels/public/file.txt
cat /levels/public/file.txt
```

Go to your Answer Sheet and write your Step 7 observation before moving to Part 2.

Part 2: First Look at Nmap

Now you'll use Nmap for the first time, the same tool you'll use again next week for vulnerability scanning. Right now, we're only looking, not attacking anything, and you're only scanning your own machine, so this is completely safe.

If you're still logged in as ana_confidential from Part 1, type exit first to return to your normal account.

```
exit
```

Step 1: Find your own VM's IP address.

Every device on a network has an IP address, think of it as its home address. Let's find yours first.

```
ip a
```

- **What you should see:** You'll see a block of text listing network interfaces. Look for a line that says something like inet 10.0.2.15, that number is your VM's IP address. Write it down, you won't need it for this activity, but you will for future labs.
- **Which line is my real IP address?**
 - You'll see more than one inet line. Ignore the one next to lo (that's the "loopback" interface, it always shows inet 127.0.0.1 and isn't a real network connection). Look instead for an interface named something like eth0 (or similar), that's your actual network address.



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Step 2: Scan your own machine.

Instead of scanning another computer, we'll scan our own VM using a special address that always means "this same machine," called localhost.

```
nmap 127.0.0.1
```

- **What you should see:** A short list of port numbers, each followed by open or closed, and a service name like ssh or http next to the open ones. Every open port is something running on your machine that could, in theory, be reached from outside.
- **What if my scan shows everything closed, or shows "filtered"?**
 - That's still a correct result, not an error. It just means your VM isn't running many services yet, or your network adapter setting is blocking the scan slightly. Record whatever you actually see, there's no single "right" number of open ports.
- **What does a real result line look like?**
 - A typical line reads something like: 22/tcp open ssh. That means port 22 is open, and it's running the SSH service. If a port instead says closed, that service simply isn't running.

Step 3: Run a more detailed scan.

The -sV option tells Nmap to also try to detect the exact version number of whatever software is running on each open port.

```
nmap -sV 127.0.0.1
```

What you should see: The same list of ports, but now with extra detail after the service name, such as a software name and version number.

Why does the version number matter?

- Knowing a service is running is one thing. Knowing its exact version is what actually matters for security, because specific versions of software are the ones that get found to have specific, documented vulnerabilities. That's exactly what next week's lesson builds on.

Go to your Answer Sheet and fill in the Part 2 Results table.

You've now tested Bell-LaPadula against a real system and gotten a first look at Nmap.

The lab report questions on your Answer Sheet tie both parts together, so take your time with them.



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Name: Pizarro, Francis Iffer C. _____ Course & Year: BS IT/ 4th _____

Part 1 Results: Bell-LaPadula Permissions Test

Fill in the "What actually happened" and "Match?" columns based on what you observed in the terminal.

Action	Bell-LaPadula says	What actually happened	Match?
Read Public file (read down)	Should be allowed	Blocked (Permission denied because ana confidential is a different user and they cannot access the file)	No
Read Confidential file (own level)	Should be allowed	Allowed The user ana confidential successfully read the file they own.	Yes
Read Secret file (read up)	Should be BLOCKED	Blocked The system denied access because the directory is restricted to its owner, ana secret .	Yes
Write to the public file (write down)	Should be BLOCKED	Blocked The system denied writing because ana confidential does not have permissions for ana public's directory.	Yes
Write to Confidential file (own level)	Should be allowed	Blocked The system denied writing because the file was created by root and is write-protected for the user.	No
Write to the secret file (write-up)	Should be allowed	Blocked The system denied writing because ana confidential does not have permissions for ana secret's directory.	No

Part 1, Step 7: Breaking the Lab

Did the write-down succeed after you loosened the Public folder's permissions to 777, even though Bell-LaPadula says it shouldn't? Write what you observed:

the write-down succeeded because changing the folder permissions to 777 allowed read, write, and execute access for all users on the system. This misconfiguration bypassed the strict write-down restrictions typically enforced by the Bell-LaPadula security model.



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Part 2 Results: Nmap Scan

Record the ports and services Nmap found on your machine.

Port	Service Name (plain scan)	Version Info (-sV scan)
nnone	theres no active services found	all 1000 ports are closed and (ignored)
n/a	all 1000 scanned ports are closed	
n/a	all 1000 scanned ports are closed	
n/a	all 1000 scanned ports are closed	
n/a	all 1000 scanned ports are closed	
n/a	all 1000 scanned ports are closed	
n/a	all 1000 scanned ports are closed	
n/a	all 1000 scanned ports are closed	
n/a	all 1000 scanned ports are closed	

Lab Report Questions

Answer these on your own, in the space provided.

1. In Part 1, which actions matched Bell-LaPadula's rules, and which didn't? Why?

the strict initial setup matched Bell-LaPadula because it blocked lower clearance users from reading higher-level files and stopped higher clearance users from writing down. However changing the folder permissions to 777 violated the model because it allowed a restricted user to write-down, breaking the "no write-down" (*-property) rule.



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2. What happened when you loosened the permissions in Step 7 of Part 1? What does this tell you about the difference between DAC (standard Linux permissions) and a formal security model like Bell-LaPadula?

loosening the permissions allowed the write-down to succeed, which completely bypassed our security goals. This shows that DAC relies on user discretion and can be easily weakened by a single bad command, whereas a formal model like bell-lapadula requires system-enforced, mandatory rules that users cannot override.

3. In Part 2, what services did Nmap find running on your machine? Were you expecting all of them?

nmap did not find any active services, reporting that all 1000 scanned ports on localhost were closed. This was expected since this is a secure isolated lab environment that isn't hostingg any public facing servers or applications.

4. How does restricting who can read a file (Part 1) differ from restricting what's visible on the network (Part 2)? Are they solving the same problem or different ones?

restricting file access controls internal data confidentiality among local users, while restricting network visibiliity reduces the attack surface visible to external threats. they solve different problems local access control versus network exposure but both are necessary for a strong, layered defense.

5. Predict: What do you think "vulnerability scanning" (next week's topic) will add on top of what Nmap showed you today?

while nmap simply maps out which ports are open and what software is running g vulnerability scanning will actually analyze those active services for known bugs and misconfigurations and security flaws. It takes us from just knowing "what is there" to knowing "what can be exploited."